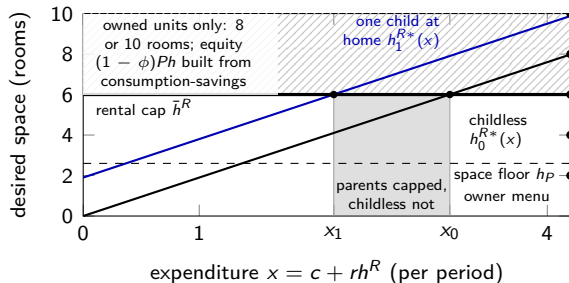


# Tenure Segmentation and the Space Floor

Within a period a renter with  $m$  children at home splits expenditure  $x = c + rh^R$  between consumption  $c$  and rental space  $h^R$  at rent  $r$  per room; the uncapped optimum is  $h_m^{R*}(x) = (1 - \alpha)x/r + \alpha mh_P$ , where  $h_P$  is the parenthood space floor and  $\alpha = 0.733$  the consumption share.

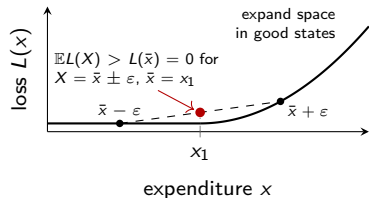


- Segmentation = floor plus cap: a parent's desired rental space crosses the cap  $\bar{h}^R = 6$  at  $x_1 = r(\bar{h}^R - \alpha h_P)/(1 - \alpha)$ , a childless household's at  $x_0 = r\bar{h}^R/(1 - \alpha) > x_1$ .
- On  $(x_1, x_0)$  only the parent is rationed; extra space exists only in the owner menu, so the family-space option depends on ownership access: liquid wealth, income this period, housing equity.
- At the benchmark ( $r = 0.141$ ,  $h_P = 2.6$ ):  $x_1 \approx 2.16$ ,  $x_0 \approx 3.16$  per four-year period; a median lower-middle-income renter spends  $x \approx 2.58$ , wants 4.9 rooms childless and 6.8 rooms as a parent.

**Takeaway.** Children raise desired space by  $\alpha h_P \approx 1.9$  rooms at any budget; the cap turns that into a tenure question for lower-middle-income renters.

# Rationed Space and Income Risk: Theory and Full-Model Check

Let  $J_1^\infty(x)$  and  $J_1^h(x)$  be a parent's value of expenditure  $x$  without and with the rental cap, and  $L = J_1^\infty - J_1^h \geq 0$  the loss from the cap:  $L = 0$  for  $x \leq x_1$  and  $L''(x_1^+) = \frac{1-\alpha}{\alpha} \frac{\lambda}{x_1 - rh_P} > 0$  ( $\lambda$ : marginal utility of expenditure). Risk around  $x_1$  costs a parent more than a childless household.



## Full-model check (fixed price, matched states, ages 22–33)

- The segmented set is populated: median-income childless renters (income state  $z = 0.78$ , zero liquid wealth), 14% of childless households, attempt probability 0.31–0.46.
- Owning does not remove the restriction: they can finance 2-, 4- or 8-room units from one period's income, yet as parents almost all rent at the cap and none buy 8 rooms or more. The shortfall is about one room (desired 6.5–7.2).
- Envelope sign confirmed, small:  $\partial G / \partial \bar{h}^R = +0.004$  utils per room ( $G$ : value gap of attempting a birth); cap 6  $\rightarrow$  8 raises their attempt probability by 2.9% (all childless +1.7%).
- No amplification: the risk effect on attempts is +50.8% with the cap and +51.2% with cap 8; zero selling cost moves it by  $-3.8$  pp. The first-order precautionary force is the space floor  $h_P$  plus the zero-borrowing floor of young childless renters.

**Takeaway.** Segmentation is real but second order here: the penalty grows with the square of the shortfall above  $x_1$ , so a lower cap or a larger parental space demand would make it bite (with cap 4 the median renter's shortfall is 1.5 instead of 0.4 expenditure units, a 12-fold penalty).